OPERATOR GUIDE · QUIKBOLUS

3D Printing a Patient Bolus on the Bambu Lab P2S

A practical, step-by-step settings + supports guide for the operator who is printing a bolus from QuikBolus output for the first time. Tuned for TPU 95A on an enclosed P2S with the textured PEI plate.

Issued 2026-05-23 · Applies to Bambu Studio 1.9+ on Bambu Lab P2S

1. Pick the right material

Direct-print bolus material has to do three things at once: stay soft enough to conform to the patient, hold dimensional accuracy through dozens of fractions, and sit close enough to water-equivalent for the planner's dose calculation to stand up. **TPU 95A** is the right default for almost every patient bolus.

MATERIAL	USE IT FOR	WHY / CAVEATS
TPU 95A *(Bambu, Polymaker PolyFlex, NinjaFlex Cheetah)*	Default direct-print bolus	Skin-flexible, conforms well, density ~1.21 g/cc — close enough to water-equivalent for most build-up applications. Hygroscopic; dry before use.
TPU 85A	Softer / more conformal bolus, very irregular anatomy	Significantly harder to print — oozes, jams, prints slower. Don't start here. Earn it on simple geometries first.
PETG or PLA *(rigid)*	Mold for casting Flexibolus or similar pourable silicone	Not for direct patient contact. Smooth interior surface = clean cast. Print the mold rigid; cast the patient-facing bolus from water-equivalent silicone.
ABS, PLA *(direct)*	—	**Avoid.** ABS warps and smells in an enclosed printer; PLA is too rigid for skin contact and not water-equivalent.

AMS warning. Bambu Lab themselves recommend running TPU from the **external spool holder**, not through the AMS. The AMS makes sharp bends that grip flexibles too hard and cause friction-jamming.

Disable AMS for the print and feed directly into the rear PTFE port.

2. Bambu Studio print profile — TPU 95A

Start from the built-in profile **"0.20 mm Standard @BBL P2S 0.4 nozzle (TPU for AMS)"** and override the values below. Save as a clinic-specific profile named *QuikBolus — TPU 95A — P2S* so the next operator gets the same settings.

SETTING	VALUE	WHY
Layer height	`0.20 mm`	0.16 for finer surface, 0.28 if speed matters. 0.20 is the comfort zone.
Wall loops	`4`	TPU at 2 walls feels noodle-like. 4 gives structural integrity to the bolus rim.
Top / bottom shells	`5`	Watertight patient-contact surface.
Sparse infill	`15–20%, Gyroid`	Gyroid keeps flexibility isotropic. Rectilinear is too stiff and prints harder lines you can feel through the skin surface.
Outer wall speed	`25 mm/s`	TPU loves slow outer walls — surface quality. Don't push past 30.
Inner wall speed	`40 mm/s`	
Infill speed	`50 mm/s`	
First-layer speed	`20 mm/s`	TPU adhesion is fussy. Slow first layer = no curling.
Nozzle temperature	`235–240 °C`	Start 235; bump 240 if extrusion stutters or inter-layer adhesion looks poor.
Bed temperature	`55 °C`	Textured PEI plate. Bump to 60 °C if first layer lifts at corners.
Retraction length	`0.5–0.8 mm`	TPU + long retraction = clog. Less is more.
Retraction speed	`30 mm/s`	Slow — don't snap the noodle back.
Bridge speed	`15 mm/s`	TPU bridges OK if slow.
Fan speed (after layer 3)	`80–100 %`	Helps definition. Layers 1–3 stay 0 % for adhesion.
Brim	`5 mm, outer-only`	TPU sometimes lifts at corners over a long print. Brim peels off cleanly post-print.
Z-hop	`OFF`	Common source of TPU jams. Turn it off.

SETTING	VALUE	WHY
Pressure advance	`0.045`	P2S default for TPU. Tune ±0.005 if you see corner bulge (too high) or thin walls (too low).

Run flow calibration once. Bambu Studio → Calibrate → Flow Dynamics + Flow Rate. 15 minutes, done once per filament brand/spool. Saves you from under-extruded thin walls on every subsequent print.

3. Supports — the part everybody gets wrong

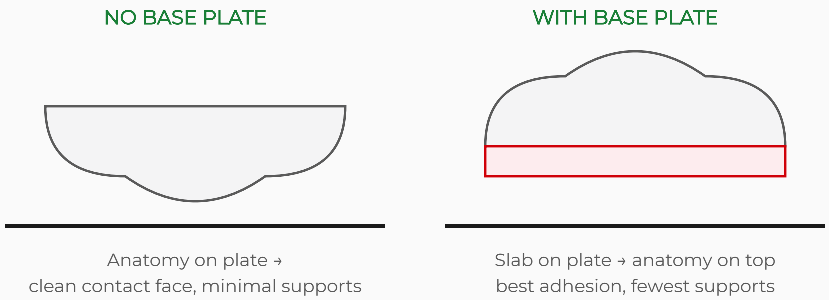
A patient bolus is anatomical, so the surface that touches the patient is rarely flat. Some overhangs are unavoidable. The trick is not "should I use supports" — the trick is **orient the part first**, then let the supports be minimal.

Step 1: Orient the part BEFORE generating supports

This is the single most important step and most people skip it. **What goes face-down on the build plate depends on whether you turned on QuikBolus's base-plate option:**

QUIKBOLUS BASE PLATE	BUILD-PLATE SIDE (DOWN)	WHY
OFF (no base plate)	Anatomy (patient-contact) face DOWN	Build plate gives the cleanest finish — the patient-contact surface gets that finish.
ON (base plate welded)	Base plate face DOWN	Flat slab against the plate = perfect first-layer adhesion + the bolus body is fully supported by the slab below, so almost no tree supports needed. The concave patient surface ends up facing UP, which prints fine (concave-upward is a self-supporting shape, not an overhang). Same surface quality, less material, less time.

Practical default. Turning the base plate ON in QuikBolus gives you the better print outcome regardless of whether you keep it on the final bolus — see section 4 for the keep-vs-trim discussion.



Both orientations work; the base-plate-down approach (right) prints faster and cleaner.

Step 2: Support settings (TPU bolus)

SETTING	VALUE	WHY
Support type	`Tree (organic)`	Less material, easier to remove than "normal". Marks TPU less.
Support style	`Snug`	Tighter contour, less waste.
Top Z gap	`0.25 mm`	**TPU is the stickiest filament there is.** Tighter than 0.25 = unremovable. 0.20 is too aggressive.
Interface layers	`2`	Smooth contact between support and part.
Interface pattern	`Rectilinear interlaced`	Comes off in clean strips.
Threshold angle	`30°`	Overhangs steeper than 30° from vertical get supported.
Support filament	`Same TPU`	Single-material print. (Dual-material PLA-on-TPU would release cleaner, but AMS+TPU isn't recommended — not worth it.)

Step 3: When in doubt, paint supports instead of auto

For an anatomical bolus, auto-generated supports often add material where you don't actually need it. Use Bambu Studio's **Paint Supports** tool to mark only the regions that overhang. The patient-contact surface (already on the plate, facing down) needs zero support. The outer dome typically needs minimal or no support — domes self-support up to ~45°.

> **Rule of thumb.** A 5–10 mm thick bolus, oriented anatomy-side-down, usually needs supports **only at the perimeter rim** where it tapers to zero thickness, and possibly under deep concavities. The bulk of the part is self-supporting. If the auto-generator wants to fill the inside of your bolus with tree trunks, something's wrong — check orientation.

4. QuikBolus's base plate option — keep or trim?

QuikBolus can weld a flat slab onto the air-side face of the bolus. You then have a choice: **leave it on as part of the final bolus**, or **use it only as a printing aid and trim it off**. Both are valid clinical workflows; pick based on what your plan calls for.

Method A — Keep it on (permanent bolus rigidity)

The base plate becomes part of the clinical bolus. It adds rigidity, gives a flat surface for handling and labeling, and lets you mount the bolus to a thermoplastic frame or indexed fixture.

- **Dosimetry note:** the base plate is in the beam path. It adds N mm of bolus material across the entire footprint — coordinate with the planner so the TPS calculation reflects the actual material.
- **Use when:** thin bolus (<5 mm) that needs stiffening; bolus that will be donned/doffed daily; you want to emboss patient ID on the flat side; the bolus mounts to a fixture.
- **Skip when:** the bolus is already self-supporting (>10 mm thick); electron treatment where the plan is to the millimeter; the planner can't re-plan to include the slab.

Method B — Use as a print scaffold, then trim it off

Print with base plate ON because that's the best print orientation (see section 3). After the print, **trim the base plate off** with a hot knife, leaving only the anatomical bolus — which now exactly matches the dosimetrist's plan.

- **Advantage:** you get the print-process benefits (flat first layer, fully supported body, clean patient surface) AND the bolus you give the patient is just the anatomy — no dose discrepancy.
- **Disadvantage:** extra post-print step; trim accuracy matters (you want to cut flush with the original bolus contour, not at an arbitrary depth).

How to trim cleanly

TOOL	BEST FOR	TECHNIQUE
Hot knife	Any thickness (gold standard)	Pre-heat tip ~200 °C; slow, single pass; let the heat do the cut. TPU melts cleanly with no tearing.
Oscillating multi-tool (Dremel)	Thick base plates (>3 mm)	Fine flush-cut blade; slow speed; light pressure. Wear safety glasses — TPU dust is fine and sticky.
Razor blade	Thin base plates (<1 mm)	Fresh blade; long single stroke through the slab; the bolus geometry below acts as a guide.
Sandpaper 220-grit	Edge finishing	After trimming, sand the cut edge smooth. Avoid sharp edges against the skin.

Pro trick: before slicing, open the QuikBolus STL in Bambu Studio and use the *Cut* tool to add a thin score line (~0.5 mm groove) at the base-plate/bolus boundary. After printing, the groove guides your hot-knife stroke and you cut on exactly the right plane.

Pick a method by scenario

BOLUS SCENARIO	METHOD	BASE-PLATE THICKNESS
< 5 mm bolus, head & neck patch	`A — keep`	1 mm (just enough to handle)
5–8 mm bolus, frequent handling	`A — keep`	1–2 mm
5–8 mm bolus, plan is exact	`B — trim`	2 mm (printing aid only)
> 8 mm bolus, self-supporting	`B — trim or skip plate`	0–2 mm
HDR cap / chest-wall flash	`A — keep`	2 mm (coordinate with planner)
Electron treatment, exact buildup	`B — trim`	2 mm (cut off post-print)

5. Bed prep — do these every time

- **Textured PEI plate** — never smooth PEI for TPU (sticks too well, you'll damage the contact surface removing it)
- Plate wiped with **isopropyl alcohol**; no glue stick, no hairspray, no Magigoo for TPU
- Filament **dried** if it's been open more than 2 weeks. Wet TPU = stringing + popping + voids. 60 °C for 6 hours in a filament dryer or food dehydrator
- AMS bypassed; filament loaded into the **rear PTFE port** directly from external spool holder
- **Flow calibration done** for this spool (Bambu Studio → Calibrate → Flow Dynamics + Flow Rate, ~15 min, once per spool)
- Bed leveled / auto-leveled within the last few prints; nozzle clean (no TPU bobble on the tip)

6. After the print — handling the finished bolus

1. **Remove from plate.** Flex the textured PEI plate gently — the bolus pops off. *Don't pry with a metal scraper*, you'll gouge the contact surface.
2. **Strip supports by hand.** Tree trunks pull off; interface layer peels. Stuck spots → flush cutters, never a knife.
3. **Inspect the patient-contact surface** under angled light. Slight plate texture transfer is fine and even desirable (small roughness helps skin adhesion). Hairline cracks or layer separation → reprint.
4. **Wash:** warm water + mild soap, rinse, isopropyl-wipe, air dry.

5. **Document** material lot, print profile name, print date, operator in the patient's chart. QuikBolus's audit record links to this.

> **Do not autoclave TPU 95A.** It deforms above ~80 °C and the bolus loses its anatomical fit. Surface disinfection only (isopropyl, quaternary ammonium wipes). Single-patient use; discard after course completion or per your institutional policy.

7. When things go wrong

SYMPTOM	LIKELY CAUSE	FIX
Print starts then jams a few millimeters in	TPU bunching at extruder gear	Lower retraction to 0.3 mm; reduce pressure advance; verify AMS bypass
Stringy gaps between perimeters	Too fast or too hot	Outer wall → 20 mm/s; nozzle −5 °C
Bolus warps off plate at hour 2	First-layer adhesion / bed too cold	Brim 5 → 8 mm; bed +5 °C
Support fused permanently to part	Z gap too small	0.20 → 0.30 mm; consider lighter interface
Patient-contact surface looks rough, layer-lined	Bridging over support gaps	Re-orient: that surface goes ON the plate, not facing up
Print time "looks insane" (12+ hours)	TPU is just slow at quality settings	This is normal. Don't chase speed; sacrifice surface quality. Plan around the duration.
Surface popping / micro-voids	Wet filament	Dry the spool 6 h @ 60 °C, reprint
Stringing between distant features	Retraction set higher than TPU likes	0.5 mm and slow (30 mm/s); some stringing on TPU is unavoidable

8. Realistic print timing

TPU is slow. Plan accordingly — chasing speed costs you surface quality and adhesion, and there is no fast TPU.

BOLUS GEOMETRY	APPROX PRINT TIME	FILAMENT USED
5×7 cm × 5 mm thick (small head & neck)	`4–6 hours`	15–25 g
10×15 cm × 7 mm (breast / chest wall patch)	`10–14 hours`	60–90 g
20×30 cm × 7 mm (full chest wall)	`18–26 hours`	180–260 g

BOLUS GEOMETRY	APPROX PRINT TIME	FILAMENT USED
20×30 cm × 10 mm (full chest wall, thick)	`24–34 hours`	240–340 g

Schedule strategy: start chest-wall prints Friday evening; treat Monday. Head & neck patches start morning before clinic close; treat next morning. Always print one spare for each course in case of damage or fit issue.

9. Alternate workflow — print a mold, cast Flexibolus

For thicker bolus, build-up where water-equivalence matters most, or anatomy that wants the very softest conformance, an alternate workflow is to **3D-print a rigid mold** (PETG or PLA) and **cast the bolus from pourable Flexibolus** or similar tissue-equivalent silicone.

Trade-offs at a glance:

	DIRECT-PRINT TPU 95A	MOLD + FLEXIBOLUS CAST
Tissue-equivalent dose	Close (~1.21 g/cc)	Very close — Flexibolus is water-equivalent by design
Skin conformance	Good	Excellent
Production time	Print only (single material change)	Print mold + cure silicone (hours added) + demold
Reuse	Single-patient, single-course	Mold lasts many casts; bolus is single-patient
Operator time	Low	Higher — pour, degas, cure, demold
Surface finish	Plate-side textured	Glass-smooth (the mold's inner surface transfers)

Use TPU 95A direct print for most cases. Reach for the mold-and-cast workflow when you need thick build-up, when water-equivalence matters more than usual (e.g., electron skin treatments, superficial photon fields), or when the anatomy is unusually irregular and you need a softer-than-TPU result.

10. Further reading & references

- **Bambu Studio 101 — Beginners Guide to Bambu Slicer Software** (YouTube): if Bambu Studio's interface is brand-new to you, watch this first. *youtube.com/watch?v=rZhy0J0mXBE*
- **Bambu Lab TPU User Guide**: official guidance on AMS bypass, PTFE routing, and material handling. *wiki.bambulab.com*
- **AAPM TG-176** (Olch et al.): dosimetric considerations for surface bolus, including discussion of bolus material density.
- **QuikBolus Users Guide**: end-to-end workflow from DICOM RT Structure Set to STL, including audit/traceability fields. *app.radiant-mpc.com/users-guides/quikbolus.pdf*

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